

BaeSlip

Team: Data Baes

Problem Statement 2 · Future Work — The future workforce
"Income without a payslip" (Airwallex)

No payslip? Your income still counts.

THE PROBLEM

People are shut out not for earning too little, but for earning illegibly.

Many students earn through casual jobs, gig apps and freelance work, but financial systems still expect stable income and recent payslips. Earnings are scattered across apps, payslips and invoices, making the full picture hard to see.

1.05M

people hold more than one job (6.9% of employed)

ABS Labour Account, Jun qtr 2026

2.4M

casual employees — 19% of all employees

ABS Working Arrangements, Aug 2025

65%

of non-ABN migrant employees were underpaid (n=5,469)

Migrant Justice Institute, 2026

\$3.18B

a year underpaid to international students (\$61M a week)

Migrant Justice Institute, 2026

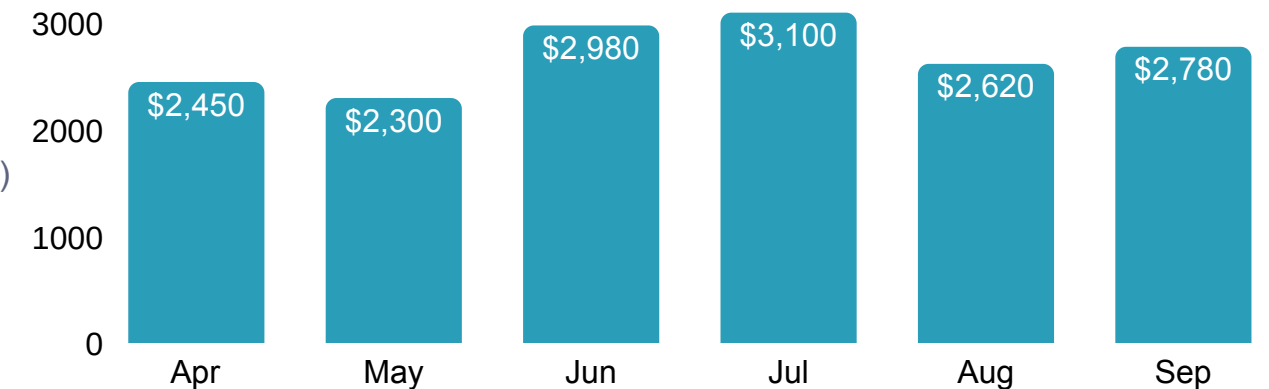
MEI, 23 · STUDENT VISA · MELBOURNE (illustrative, from our mock data)

\$16,230

attested in 6 months from 3 sources (tiers A–C)

\$2,300–3,100

a month · median \$2,700



4 different income:

A

Café casual

B

Delivery app

C

Overseas design

D

Cash tutoring

Today "Send recent payslips."

Evidence is scattered across apps, payslips and invoices. No single document shows the full picture.

With BaeSlip

Scan QR for live status: valid, expired or revoked. 6 months of income · evidence level per source.

OUR SOLUTION

A worker-controlled, scannable income proof — and an open standard any bank can issue.

BaeSlip combines irregular earnings in one timeline, labels evidence strength and issues a signed proof for live QR checks. Private checks help workers spot possible issues.

1 Evidence tiers A–D

- A Government-reported source
- B payer-confirmed
- C matched to bank deposit
- D self-reported.

Cash shown honestly, never inflated into the proof.

2 Signed, live QR proof

Ed25519-signed. Scan shows Valid / Expired / Revoked in real time. Change one number and verification fails. Worker revokes anytime.

3 Privacy by design

Proof carries income only. Hours, visa status, transactions and check results can never appear. Worker previews both lists first.

4 Worker-only checks

Min. wage, delivery pay floor, late super, 48-hour visa cap. Worded "may", with next steps. AI explains in her language; code does the maths.

A standard, not just an app: proof format, tiers and never-share list are open; BaeSlip is one issuer. Who pays: free for workers and landlords. Hypothesis: lenders, who must verify income (ASIC RG 209), pay per check.

Tech & Progress

Built, tested and working end to end — on mock data for one illustrative worker.

385 tests passing

197 backend · 188 frontend

8 API endpoints

issue · verify · revoke · keys · ...

4 private checks

all firing on Mei's data

4 screens

app · add data · verify · spec

 phone demo

scan → Valid → Revoked

Tech stack / tools

Backend

Python 3.11 · FastAPI · Pydantic v2

Crypto

Ed25519 signatures via cryptography · canonical JSON (sorted keys, no whitespace, UTF-8)

Frontend

React 19 · Vite · React Router · qrcode.react

AI

Claude API (Anthropic) — plain-language explanations only; template fallback

Standard

JSON Schema 2020-12 generated from Pydantic · public keys at /.well-known

Storage

JSON files (demo) · amounts as integer cents

Quality

pytest · Vitest · Testing Library · GitHub Actions CI · contract check

Built by

test-first, with parallel AI coding agents (Claude Code, Codex)

Data flow — from evidence to a signed proof

1

Collect bank deposits via CDR · myGov income statement · platform earnings statement · client invoices · rosters · super statement · cash she declares

2

Match each income is matched to a bank deposit: amount within \$1 (2% for foreign currency), paid within 3 days, payer name in the description

3

Tier a record takes its best evidence: STP-checked → A · platform or payer-confirmed → B · bank match only → C · declared only → D. A source takes its weakest record

4

Process integer cents; monthly totals from A–C only → range, median, coverage. Rosters and super feed private checks only. Proof is Ed25519-signed

Is Mei being paid fairly? 4 Private alerts

R1

Minimum wage
\$221 may be short
\$661 due · \$440 paid

R2

Delivery floor
\$31.70 may be short
\$281.70 due · \$250 paid

R3

Payday Super
Late arrived 28 Sep
due 25 Sep

R4

Visa hours
50 h of 48 allowed
warned before it happens

How it works

Real screens from our running app — the worker shares, the worker revokes.

1

Add data

My income [Add your data](#)

[The standard](#)

Add your data

Demo: nothing is uploaded or processed. Mei's data is already loaded.

Bank account

Connect with Consumer Data Right (CDR). You approve the share at your bank; no file needed.

Tier C: Matched bank deposit

Connect bank Not added

myGov income statement (STP)

Download it from myGov (ATO online)

2

See the summary

My income [Add your data](#)

[The standard](#)

Mei's income proof

Illustrative demo: Mei, a student-visa holder with four income sources, April to September 2026.

Income summary

Monthly income range
\$2,300-\$3,100 per month (AUD)

Median month
\$2,700

Coverage
6 of 6 months with data

Low \$2,300 Median High \$3,100

Hospitality (casual) A Gov-reported

Food delivery platform

3

Private alert

Checks for you only

These checks are private to you. They are not shared with landlords or lenders, and they never appear on an income proof or its verify page.

Minimum hourly rate

Pay may be below the minimum rate

Needs review

Difference \$221.00

Expected \$661.00

Paid \$440.00

1-14 Sep 2026

Explain in my language

Details

4

Preview sharing

Share a rental attestation

Hide preview

Will be shared

Your display name (Mei L.)

Income source names, types and evidence tiers

Monthly income range

Median monthly income

Months with data (coverage)

Period the proof covers

Issuer, issue date and expiry date

Will not be shared

Hours and shifts (never shared)

Visa status (never shared)

Individual transactions and bank deposits (never shared)

Integrity check results (for example, possible underpayment) (never shared)

Income for each month (not in

5

Landlord: Valid

Verify income attestation

Attestation ID: att_h-KL-oXNXP0

Valid

Income for Mei L.

1 Apr – 30 Sep 2026

Monthly income range: \$2,300-\$3,100

Monthly median: \$2,700

Low \$2,300 Median \$2,700 High \$3,100

6 of 6 months with data

Sources

6

Mei revokes

Verify income attestation

Attestation ID: att_h-KL-oXNXP0

Revoked

The person who shared this attestation has withdrawn it.

Revoked at 2026-09-30 12:03

Bank (CDR), myGov STP, platform, invoices.

\$2,300–3,100 · median \$2,700 · 6/6.

\$661 due, \$440 paid: \$221 gap.

Shared vs never shared.

Scan shows income only.

Same link: Revoked.

Architecture

```
graph LR
    Evidence["Evidence  
A myGov · B platform / payer  
C bank · D self"] --> TS["Timeline + tiers  
matching · summary"]
    WorkerApp["Worker app  
React · /app"] -- API --> Checks["Checks R1-R4  
worker only"]
    Checks --> Explain["Explain · Claude API  
rewords only · template fallback"]
    TS --> PB["Proof builder  
whitelist · scope"]
    PB --> ES["Ed25519 signer  
canonical JSON"]
    ES --> PS["Proof store  
signed proofs · revoked_at"]
    PS -- verify --> Landlord["Landlord  
verify page /v/:id · QR  
(lender: spec only)"]
    PS -- live status --> Landlord
    PS -- "revoke (no re-signing)" --> WorkerApp
    PublicKeys["Public keys  
/.well-known"] -.->|verify independently| Landlord
    PublicKeys -.->|same key pair| ES
```

Legend: → data flow, - -> public key, → worker revokes

Notes:
- never enters the proof (between Checks and Proof builder)
- Only income leaves. Checks never reach the proof. (between Checks and Explain)

Why not bank statements?

Statements expose every purchase and are easy to edit. Employer letters cover one job. Lender tools see one account and belong to the lender. **BaeSlip: multi-source, evidence-tagged, signed, worker-controlled.**

Scan for GitHub

https://github.com/JackLee083/DataBaes_BaeSlip